

Acoustics of the Gabriel's Horn

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Introduction

One time, I made joke with friends, that I was given task to clean Gabriel's horn out of rust in order to play it. It brings question, how it will actually sound like? So here is the analysis.

Theoretical setup

I model the Gabriel's horn with Webster's horn equation (??). This differential equation describes the pressure inside the horn with cross-sectional area $S(x)$ and assumes that waves are approximately plane. For derivation see appendix.

$$\frac{1}{S(x)} \frac{\partial}{\partial x} \left(S(x) \frac{\partial p}{\partial x} \right) = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \quad (1)$$

Where, $p(x, t)$ is the pressure in the horn and c is speed of the sound.

Substituting cross-sectional area of horn $S(x) = \frac{\pi}{x^2}$, we get following equation.

$$\frac{x^2}{\pi} \frac{\partial}{\partial x} \left(\frac{\pi}{x^2} \frac{\partial p}{\partial x} \right) = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \quad (2)$$

Now we need focus on waves that exhibit the harmonic oscilations, so we make substitution of $p(x, t) = \psi(x) \cdot e^{-i\omega t}$.

$$\frac{x^2}{\pi} \frac{\partial}{\partial x} \left(\frac{\pi}{x^2} \frac{\partial \psi}{\partial x} e^{-i\omega t} \right) = - \left(\frac{\omega}{c} \right)^2 \psi(x). \quad (3)$$

We introduce new variable $k = \omega/c$, for simplification of notation.

...

$$x^2 \left(\frac{\partial}{\partial x} \left(\frac{1}{x^2} \right) \frac{\partial \psi}{\partial x} + \frac{1}{x^2} \frac{\partial^2 \psi}{\partial x^2} \right) \quad (4)$$

...

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{2}{x} \frac{\partial \psi}{\partial x} + k^2 \psi = 0. \quad (5)$$

We impose initional condition such as that at throat ($x = 1$) is open end, $\frac{\partial \psi}{\partial x} = 0$, and closed end at $x = L$ with $\psi L, t = 0$.

The system that we described can be solved by Bessel equations, with some ...

We arrive at

$$k(L - 1) + \arctan(k) = n\pi \tag{6}$$

We solve following using fixed point iterations algorithm.